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Rock Imaging Reveals Flow Anisotropy in Mega-Porous Carbonates

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Abstract

Dissolution secondary porosity represented in the core as vugs within carbonates, represents enhanced fluid capacity and filtration potential. This paper outlines the results of the reservoir characterization study on such a pore system from one of the Jurassic formations in Saudi Arabia. Refinement of permeability values were investigated by employing whole-core (WC) computed tomography (CT), 3D X-ray microscopy or micro-computed tomography (micro-CT) and digital rock analysis computational workflows. Studied limestones contained diagenetic dissolution fabrics displaying comparatively higher porosities in oolitic, mixed skeletal peloidal, and fragmented cladocoropsis facies.

Sample selection was based on the screening of associated WC images and core viewing. Associated criteria were focused on the availability of plugs measured properties, and where available WC laboratory petrophysical properties. The digital rock analysis workflow included: 1) scanning of the WC samples by CT at voxel size of 200 µm; 2) micro-CT at a voxel size of 30 µm; 3) high-resolution imaging at a voxel size of 35 µm; 4) segmentation of the tomograms to identify pores and grain fabrics and 5) building the integrated pore network models and calculation of petrophysical properties. Digital analysis of air permeability of both macro- and microporous pore space was achieved through the Stokes equation in three orthogonal directions illustrating flow heterogeneity.

Twelve whole-core sections were imaged with high resolution micro-CT as they represented the most vuggy intervals seen during the core screening. This allowed an adequate section of the WC to be imaged and represented a much larger sample size than the corresponding plug set. Vugs to matrix hydraulic connectivity is a feature of the pore system that could only be extracted from micro-CT images. Permeability simulation results yielded new porosity/permeability trends and permeability anisotropy that were further utilized to update 3D geological models. This study was able to supply evidence of how the flow anisotropy could be resolved using micro-CT for high-end non-matrix porosity carbonates and facilitated the understanding of petrophysical differences between plug and WC components, as well as the matrix and non-matrix components. The workflow used, not only predicted rock properties, but also the spatial distribution of the rock features and pore network assemblages causing these properties.

Due to the well-known issues with conventional laboratory measurements on vugular WC, this study demonstrates that reliable porosity and permeability values can be obtained using digital analysis such as micro-CT or CT-scanning on WC at the finest resolution possible. The data infer that digital rock analysis

is the only method that yields flow anisotropy in the aforementioned vugular rocks. The results improve the understanding of storage capacity and recovery mechanisms of the studied formation.

Introduction

The characterization of pore systems in limestones is crucial for understanding their fluid capacity and flow potential. In the Jurassic formation of Saudi Arabia, dissolution secondary porosity plays a significant role in determining the reservoir's properties. The presence of diagenetic fabrics, exhibited higher porosities in oolitic, mixed skeletal peloidal, and fragmented cladocoropsis facies, adding complexity to the pore system.

As pre-work for this high-end dissolution porosity study, a review of historical WC analyses was conducted to refine permeability values and improve the understanding of vugular carbonate reservoirs. WC analyses have been touted in the past as the best analysis technique to capture the reservoir porosity and permeability heterogeneity, especially in zones where the reservoir is very vuggy. The major occurrences of vugs in carbonate formations in the middle east have been highlighted by many authors, but summarized more recently by Cross & Burchette (2024). Internally we categorized our carbonate into three main incidences: limestone containing diagenetic improvement of porosity in the form of dissolution in oolitic, mixed skeletal peloidal, fragmented cladocoropsis or cladocoropsis lithofacies; dolomitized (fabric preserved) leached cladocoropsis; and dolostone (non-fabric preserved) exhibiting mostly sucrosic textures. Examples of the dissolution that we observe in cores are given in Figure 1.

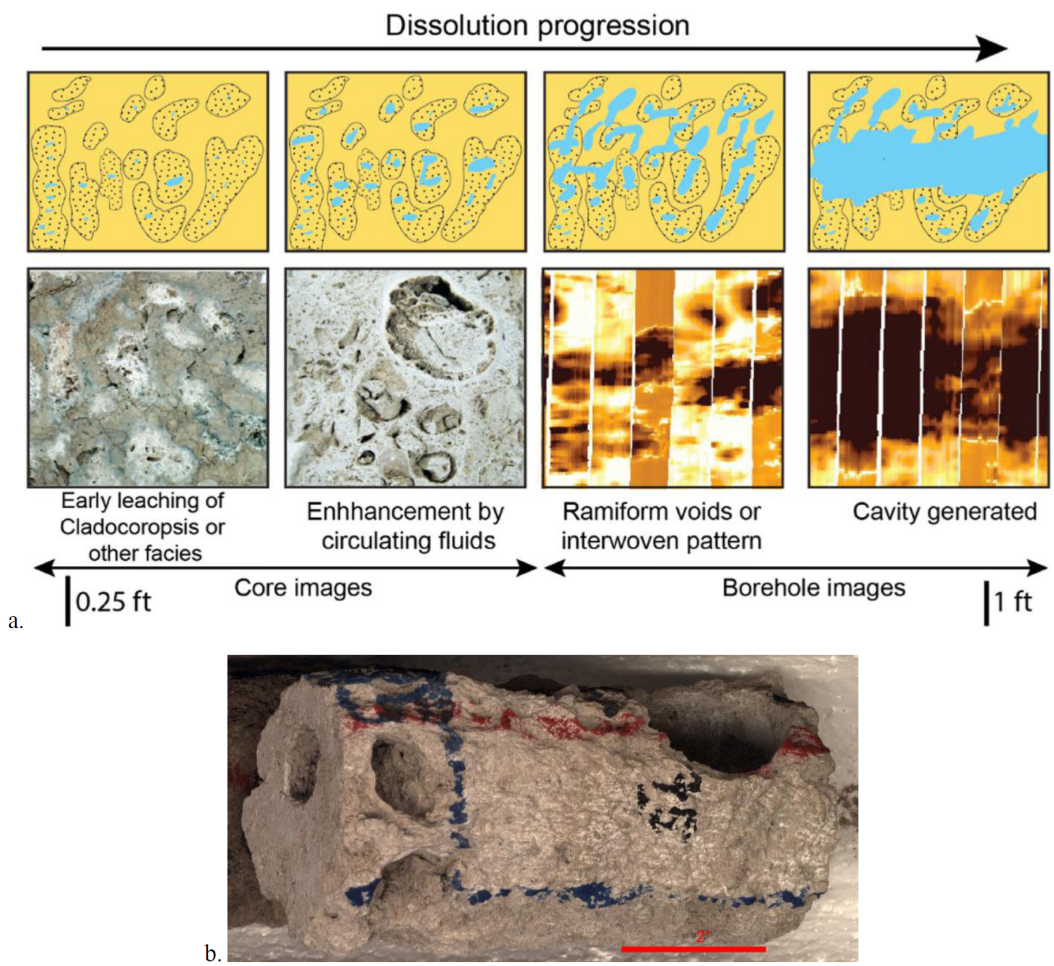

Figure 1—Example of the progression of dissolution features observed in core and as scale increases are determined from logs: a. progression observed, b. whole core with routine core analysis (RCA) plugs extracted. RCA plugs cannot capture porosity heterogeneity attributed to larger scale dissolution features.

However, the review of the published literature revealed that laboratory-based WC analyses, although robust for porosity measurements, can yield erroneous permeability values Laboratory limitations with deep mud invaded vuggy core samples (Honarpour *et al*, 2003) highlighted the need for alternative methods to accurately characterize the pore system and derive a realistic subsurface permeability. Authors such as Mitchel-Tapping (1982) have stressed that there are also some other obvious issues with cleaning that would affect subsequent analysis measurements. Given all these laboratory related experimental issues, recent advances in CT and digital rock analysis have enabled the development of computational workflows that can process WC data at high resolution, avoiding the laboratory pitfalls (Wildenschild and Sheppard, 2013; Blunt, 2017; Zhao et al., 2019; Santos et al., 2020; Prodanovic et al., 2023).

Vuggy zones observed in the core have been investigated in the past as they usually coincide with increased productivity within the reservoir and are often associated with reservoir "Super-permeability" zones or total losses while drilling. This study aims to establish a practice in enhancing core analyses, leveraging digital technologies to improve the characterization of complex pore systems and improve the acceptance of digitally simulated permeability estimates.

Challenges Evaluating High-End Dissolution Porosity

Internally we have a large volume of studies highlight the importance of WC analysis in characterizing vuggy carbonate reservoirs going back to the 50's. In light of more recent publications such as Honarpour *et al* (2003), the historical reasoning to utilized larger volumes of core were well founded, but flawed when it came to the actual laboratory testing. Mud invasion of CaCO3 particles, as well as a number of other weighting agents in the mud, are the main cause of the mis-matches we are observing in our data sets.

Observations such as zones of poor core recovery are often encountered in intervals of very vuggy rock or in crestal locations where dissolution was more prominent, which bring about challenges when evaluating poroperm using conventional core analysis techniques. Core that remained intact during the retrieval process which are friable are still complete enough to undergo CT scanning. Historically, laboratory WC analysis were used to provide more representative porosity and permeability values in these complex reservoirs.

Previous core-based studies have demonstrated that very vuggy cores are often associated with moldic porosity found within leached Cladocoropsis. In such cases, it is recommended that these zones be evaluated using whole core analysis rather than relying on 1″ or 1.5″ plugs. This is because plugs may not capture the full range of porosity and permeability variability present in these vuggy zones. The difference in porosity between plugs and WC at the same depth could be up to 17pu, emphasizing the need for WC analysis in these complex reservoirs.

A more recent review of the porosity data from both WC and plug samples revealed a notable discrepancy, with up to 7pu difference in porosity values between the two datasets in the reservoir section (Figure 2). This significant mismatch can be attributed to the distinction between matrix and non-matrix components of porosity. Specifically, our analysis showed that WC samples exhibited substantially higher porosity values compared to the corresponding plug samples in vuggy intervals. This is consistent with expectations, as the larger volume of rock represented by the whole core samples is more likely to capture the larger-scale porosity features, such as vugs, that are often missed by the smaller plug samples. This observation highlights the importance of considering the scale of measurement when evaluating porosity in complex reservoirs.

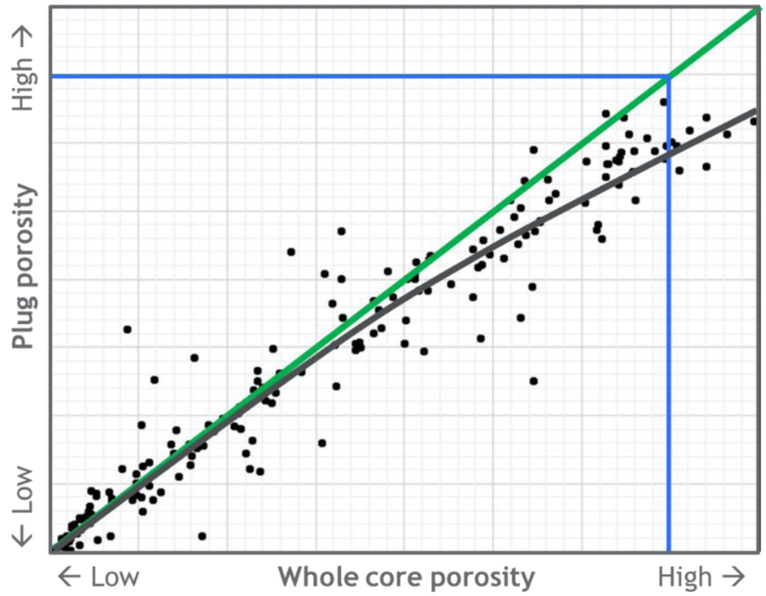

Figure 2—Cross-plot illustrating comparison of porosity derived from whole core and plugs analyses from internal legacy core analysis reports. The mismatch can be attributed to the distinction between matrix and non-matrix components of porosity

Despite the benefits of whole core analysis, the programs implemented in the past have been plagued by technical issues that have brought into question the accuracy of the permeability results. A review of the whole core analysis practices of the time reveals several key issues that contributed to these problems, including inadequate core handling and preservation techniques, insufficient sampling protocols, limited understanding of the reservoir geology and petrophysics and technical limitations of the analysis equipment and techniques, which may have limited the accuracy and precision of these results.

To improve the accuracy and reliability of WC analysis results, it is essential to address these issues and develop improved methods and protocols for characterizing vuggy carbonate reservoirs. This includes the use of advanced equipment and techniques, such as computed tomography (CT) and digital rock analysis, as well as the development of more robust sampling protocols and quality control procedures.

As highlighted by Honarpour et al. (2003), there is a growing recognition of the limitations of laboratory whole core measurements of porosity and permeability. Instead, digital rock analysis and CT-scans are emerging as a more reliable and accurate method for characterizing vuggy reservoirs.

In contrast to laboratory examination of cores, digital analysis using CT-scans offers a number of advantages. By creating a detailed, three-dimensional image of the core, CT-scans allow for the virtual removal of invading components, such as mud solids, from the analysis. This enables the delivery of a reliable porosity and permeability measurements, which is essential for accurate reservoir characterization. The use of digital rocks and CT-scans is therefore becoming increasingly important in the analysis of vuggy carbonate reservoirs, as it provides a more accurate and reliable method for measuring porosity and permeability.

The benefits of digital analysis are clear, and it is likely that this approach will become the standard method for analyzing vuggy reservoirs in the future.

Samples Selection

The selection of samples for this study was a critical step in ensuring that the results would be representative of the formation's properties. To achieve this, a thorough screening of associated core images and core layout was conducted to identify the most suitable sections for analysis. Additional criteria were also considered, including the availability of corresponding plug-measured properties, to provide a comprehensive dataset for comparison.

A total of 12 WC sections ranging between 3 to 8″ and 2.5″ in diameter were chosen for this study, representing a range of vuggy and non-vuggy intervals. The vuggy intervals were selected as they are characteristic of the reservoir's complex pore system, and their analysis would provide valuable insights into the performance of CT-scanning in capturing the digital porosity and permeability. The inclusion of non-vuggy samples, on the other hand, allowed for the testing of CT performance in the digital porosity computation with different sample types, ensuring that the results would be applicable to a wide range of conditions.

The chosen sections were also selected based on the location of corresponding RCA plugs taken after the WC lab analysis, and taken towards the end of the WC sections. This allowed for a decent section of the WC to be CT-scanned, providing a much larger sample size that is closer to the WC than the corresponding plugs. By analyzing these larger samples, the study aimed to capture the heterogeneity of the reservoir and provide a more accurate representation of its properties.

The sample selection process was designed to ensure that the results would be representative of the reservoir's variability, and that the comparison between CT-scanned whole core samples and plug-measured properties would be meaningful.

The Digital Rock Workflow and Results

A comprehensive digital rock physics workflow was employed to characterize the internal fabric of 12 selected WC sections. The workflow involved a multi-resolution X-ray tomographic imaging approach, combined with voxel-based numerical simulations, to produce 3D digital rock models of the rock samples. The workflow consisted of four main steps as per below.

1. **Scoping scanning**: WC samples were scanned at a resolution of 200 μm voxel size, followed by high-resolution scanning at 25-30 μm voxel size to examine the samples' heterogeneity and pore structure (Figure 3).
2. **Sub-volume scanning**: Sub-volumes were selected and scanned at a resolution of 3-5 μm voxel size to characterize micro-heterogeneity (Figure 4, A).
3. **Pore network characterization**: The micro-porous structures with macropores (vugs) were delineated and composed into the total pore network.
4. **Petrophysical properties calculation**: The pore structure and petrophysical properties, including porosity, pore size distribution, and absolute permeability in three orthogonal directions (parallel and perpendicular to whole-core axis), were simulated and corresponding datasheets were generated at the micro-scale (Figure 4, B-F) and for composite samples (Figures 5 and 6).

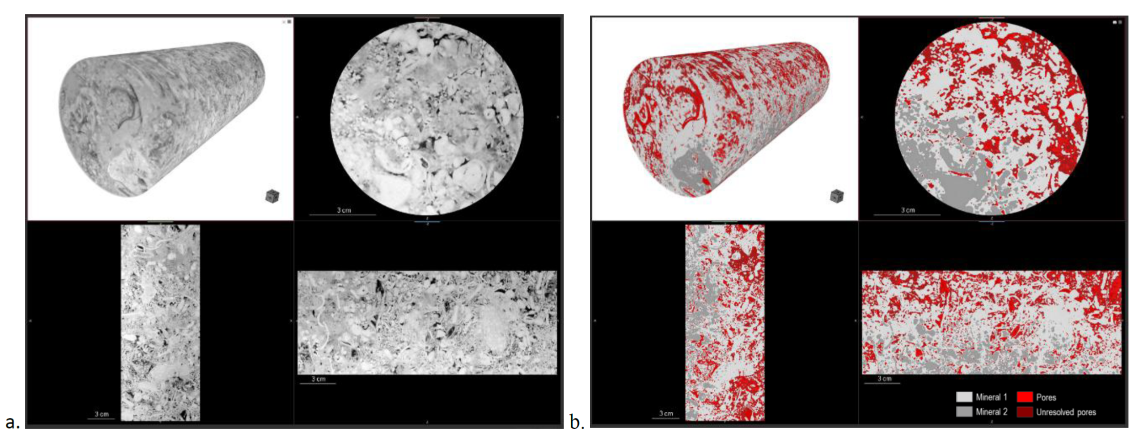

Figure 3—Macro-CT scanning and analysis results: a. raw gray-scale tomograms and b. segmented tomogram, 28 μm voxel size.

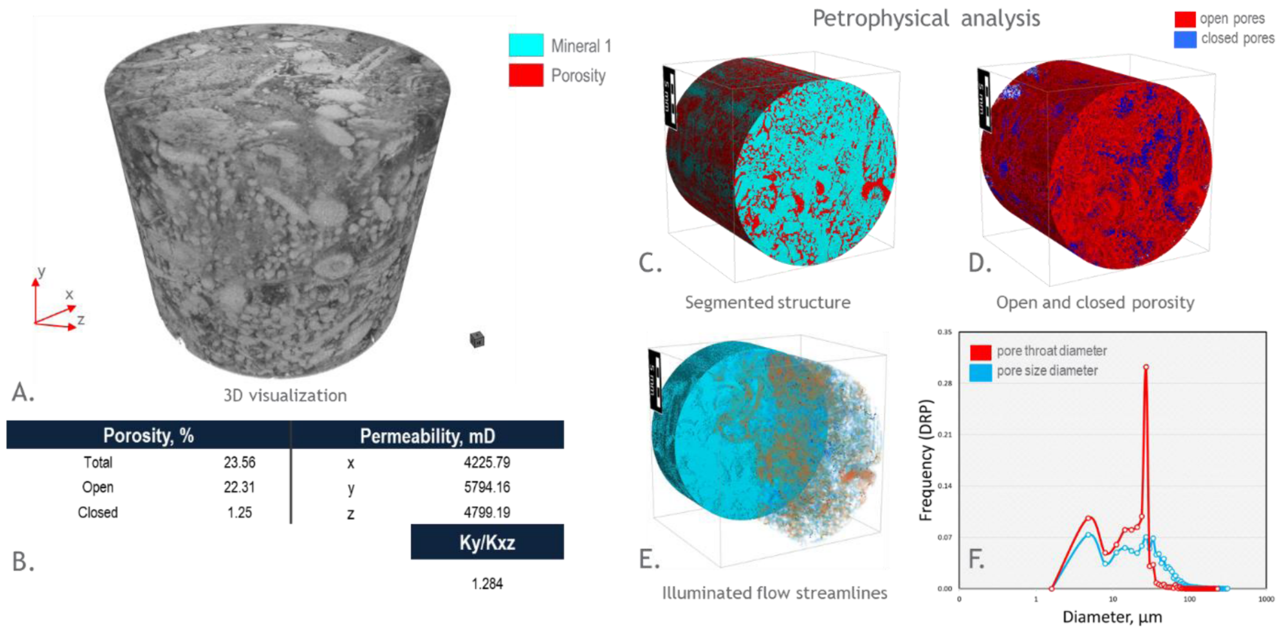

Figure 4—Micro-CT scanning and analysis results: A. raw gray-scale tomogram, B. calculated properties summary, C. segmented tomogram, D. open and closed porosity structure, E. rock structure and flow lines, F. calculated pore throat diameters; 3 µm voxel size.

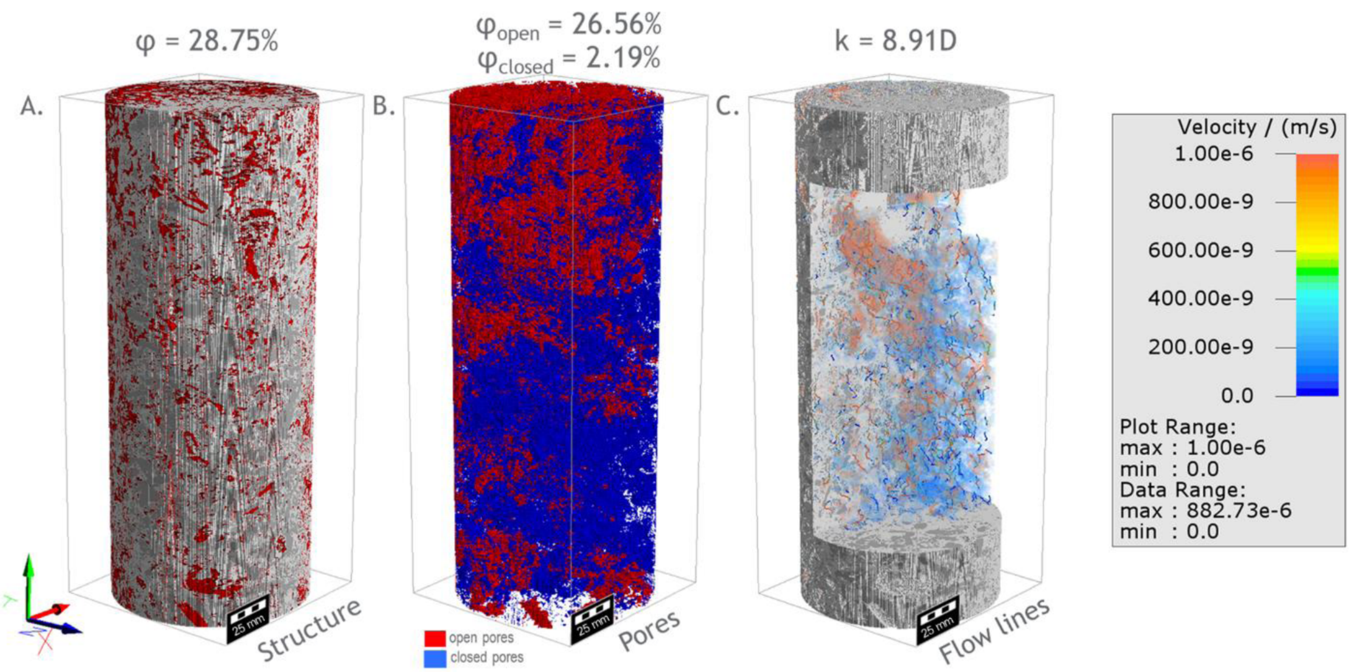

Figure 5—An example of composite digital rock model: A. segmented structure, B. open and closed porosity structure, C. illuminated flow lines.

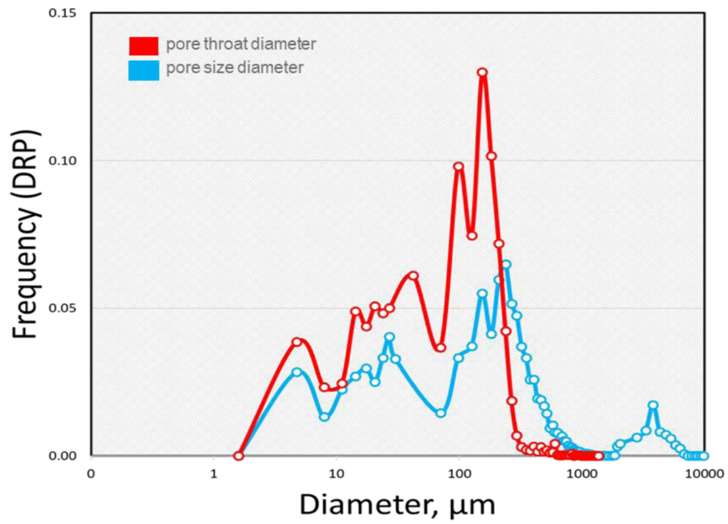

Figure 6—Pore sizes and pore throat sizes distributions derived from the composite digital rock model illustrated on the Figure 5.

CT Scanning

CT scanning of the selected vugular sections of the WC was performed in-house by Saudi Aramco's imaging laboratory. The scanning parameters included a power of 160kV at a 25W voltage, using a HE2 filter with a transmission of 25% and an exposure time of 5.2 seconds.

Image Processing and Analysis

The pore space was segmented using a combination of thresholding and watershed transform applied to the gradient 3D magnitude volume of the sample's tomography. The pores were separated using a geometrical watershed transform algorithm, allowing for the isolation and analysis of the volume and geometries of the pore bodies. The pore throats were then extracted, and a pore size distribution was computed. Absolute permeability of both macro- and microporous pore space was calculated through the solution of the Stokes equation for absolute permeability (Toelke, 2010).

Digital Rock Physics Upscaling

The digital rock physics upscaling methodology utilized the Navier-Stokes-Brinkman equation to simulate fluid flow in resolved and unresolved pores, represented with permeable material of 5 mD permeability. The composite model employed an adaptive grid, rather than a regular grid, with varying cell sizes depending on the pore geometry complexity.

Comparison with Laboratory Whole Core Analysis

The results of the CT-scans analysis were compared to the corresponding laboratory WC analysis to make a like-to-like comparison (Figure 7). The comparison revealed a minor discrepancy in porosity, but a significant difference in permeability measurements. This highlights the value of micro-CT scanning WC sections, as the highly permeable sections are often underrepresented in laboratory measurements.

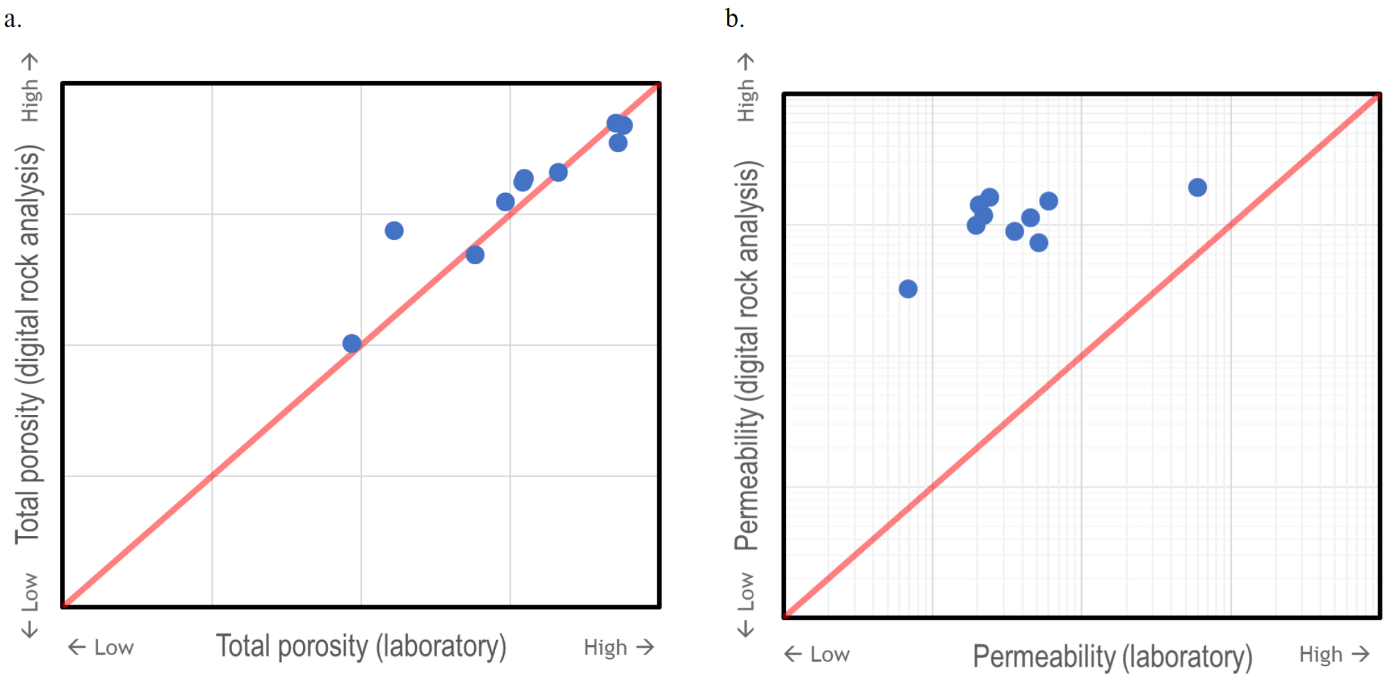

Figure 7—Comparison of laboratory- and digital rock-derived properties relating to the same whole core samples: a. total porosity and b. absolute permeability.

Conclusions

This study has demonstrated the advancements and limitations of laboratory-based WC permeability measurements in vuggy rocks, highlighting the need for alternative methods to accurately characterize these complex reservoirs. The laboratory-based measurements are often compromised by factors such as inadequate cleaning practices, core damage, and the inability to remove mud solids and filtrate invasion, which can significantly reduce porosity and permeability.

The review of data has shown that laboratory porosity measurements from WC are fairly robust in sections with isolated vugs. However, when the core displays vugs or larger interconnected vugs, the porosity difference between plugs and WC can reach 5 p.u. with average deviation of 2 p.u. This highlights the importance of using digital rock analysis to characterize the pore system and extract features such as vugs to matrix hydraulic connectivity, which cannot be obtained from laboratory measurements alone.

The developed digital rock workflow has enabled the analysis of micro-CT images to extract new porosity/permeability trends and permeability anisotropy, which have been utilized to update the 3D geological models. This study has provided evidence of how flow anisotropy can be resolved using micro-CT, and has improved our understanding of the petrophysical differences between plug and WC components, as well as the matrix and non-matrix components. The workflow has not only been used to predict rock properties but also to characterize the spatial distribution of rock features and pore network assemblages that cause these properties.

Based on the findings of this study, the following recommendations are made for future studies of vugular rock:

1. To obtain reliable porosity and permeability values, digital analysis of micro-CT scanned WC at the highest resolution possible should be used. This will allow for better characterization of porosity types and permeability.
2. The digital analysis should be conducted in a way that any WC mud invasion (or deposited salt) can be digitally removed from the whole-core digital rock model, prior to the analysis of porosity and

permeability. This will ensure that the measurements are accurate and representative of the reservoir properties.

By adopting these recommendations, future studies can improve our understanding of vugular rock and provide more accurate characterizations of these complex reservoirs. The use of digital rock analysis and micro-CT imaging has the potential to greatly improve the field of reservoir characterization, and this study has demonstrated the value of these techniques in refining our understanding of petrophysical properties of vugular rock.

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